

A Consent-First Personal AI Mentor Ecosystem: A Multi-Agent Framework for Lifelong Learning, Wellbeing, and Human Flourishing

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August 2026

Abstract

Many of the most damaging trajectories in human life (childhood distress that goes unnoticed, drift into addiction, inconsistent learning, and loss of purpose) share a common root: no one was paying sustained, caring attention at the moments when small interventions would have mattered most. Advances in large language model (LLM) agents make sustained, personalized attention technically feasible for the first time, but the obvious architecture, pervasive institutional monitoring, is both ethically unacceptable and, we argue, *counterproductive*: coerced observation destroys the trust on which mentorship, disclosure, and behavior change depend. This paper proposes a consent-first alternative: a multi-agent ecosystem in which every individual owns a Personal Mentor Agent (PMA) that supports their learning, health habits, and long-term goals, and institutions operate Institutional Coordinator Agents (ICAs) that can interact with a person's PMA only through explicit, revocable, scoped consent. We present the architecture, including an agent-to-agent consent protocol, a personal data vault with local-first storage, and human-in-the-loop escalation for safety-critical events, together with illustrative scenarios in education, healthcare, homes, and workplaces. We position this as a framework and research-agenda paper: rather than reporting empirical results, we specify a concrete evaluation plan with falsifiable hypotheses and measurable outcomes, and we discuss the ethical boundaries that distinguish a mentorship ecosystem from a surveillance apparatus. Our central claim is that consent is not a constraint on such systems but the mechanism that makes them effective.

Keywords: Personal AI agents, multi-agent systems, LLM agents, consent architecture, AI mentorship, education technology, digital wellbeing, privacy by design, human-centered AI

1. Introduction

Consider three familiar failures of modern life. A child experiences something painful, such as bullying, a family conflict, or a humiliation at school, and because no adult notices or asks, the experience calcifies into anxiety or depression that shapes decades of their life. A teenager, disconnected and in quiet pain, experiments with substances despite years of warnings, because the warnings addressed information he already had rather than the loneliness that drove him. An ambitious learner in an age of abundant knowledge jumps from topic to topic, never receiving the consistent, adaptive guidance that turns interest into mastery.

None of these failures stems from a lack of information. They stem from a lack of *sustained, personalized, caring attention*: the kind a devoted mentor, tutor, or attentive parent provides, and which

has never been scalable to every human being. Recent progress in large language model (LLM) agents changes this calculus: for the first time, software can hold long-running, context-rich, adaptive conversations; remember an individual's goals and history; and coordinate with other software agents on the individual's behalf [1], [2].

The temptation, when imagining such systems at societal scale, is a top-down architecture: cameras and sensors everywhere, institutional AI systems that observe everyone continuously, mandatory participation, and automated escalation to authorities. We argue this architecture fails on its own terms, independent of its ethical problems. Mentorship, honest disclosure, and durable behavior change all depend on trust, and trust cannot be established under compulsion [3]. A child interrogated daily by a mandatory system learns to hide; a monitored adult learns to perform. Surveillance produces compliance signals, not flourishing.

This paper develops the alternative. We propose a **consent-first personal AI mentor ecosystem** built on two agent classes. Each person owns a **Personal Mentor Agent (PMA)**: a long-lived AI companion, controlled by the individual (or, for minors, jointly with parents or guardians), that supports learning, health habits, emotional check-ins, and long-term goals. Organizations: schools, clinics, workplaces: operate **Institutional Coordinator Agents (ICAs)** that manage institutional functions and may exchange information with a person's PMA *only* through explicit, scoped, revocable consent grants. The individual, not the institution, is the root of authority over their own data and their own agent.

The contributions of this paper are:

1. A design analysis showing why consent-first architectures dominate surveillance architectures even on effectiveness grounds, drawing on evidence from developmental psychology, addiction research, and intelligent tutoring systems.
2. A reference architecture for a two-tier multi-agent ecosystem (PMAs and ICAs), including an agent-to-agent consent protocol, a local-first personal data vault, and human-in-the-loop escalation paths for safety-critical events.
3. Illustrative cross-domain scenarios in education, healthcare, homes, and workplaces that preserve the benefits commonly claimed for pervasive smart environments while eliminating coercion.
4. A concrete evaluation plan with falsifiable hypotheses, proposed metrics, and study designs, offered as a research agenda for the community. We explicitly do not report empirical results in this paper.

The remainder of the paper is organized as follows. Section 2 states the human outcomes the system is designed to serve and derives design principles from them. Section 3 reviews related work. Section 4 contrasts the surveillance architecture with the consent-first architecture and explains why the latter is more effective. Section 5 presents the proposed architecture in detail. Section 6 walks through illustrative scenarios. Section 7 specifies the evaluation plan. Section 8 discusses ethical considerations and limitations, and Section 9 concludes.

2. Motivation and Design Goals

2.1 Target Human Outcomes

The system is designed around five outcomes, stated in human rather than technical terms:

- **O1: Early emotional support.** Distress in children and adults is noticed early, because the person has a trusted companion they choose to confide in, and met with support and, where appropriate, connection to human help, before it compounds into long-term harm.
- **O2: Prevention through connection.** The drivers of substance abuse and self-destructive behavior (isolation, untreated pain, absence of purpose) are addressed at the root. Decades of research indicate that information campaigns alone do not prevent addiction; connection, mattering, and early support do [4], [5].
- **O3: Consistent mastery.** Learners of any age receive adaptive, persistent guidance: a curriculum shaped to their interests, daily practice appropriately drilled and spaced, and encouragement calibrated to their state: the demonstrated benefits of one-on-one tutoring [6] made available to everyone.
- **O4: Healthy daily life.** Sleep, nutrition, movement, and routine are supported through gentle, personalized nudges the person has asked for, in the tradition of positive behavioral scaffolding rather than enforcement.
- **O5: Purpose and economic dignity.** By helping people discover and develop genuine competence, the ecosystem widens paths to meaningful work, contributing over time to reduced poverty and exclusion.

2.2 Design Principles

From these outcomes and from the effectiveness analysis in Section 4, we derive five principles that constrain every architectural decision:

- **P1: Individual ownership.** The PMA belongs to the person. They can inspect what it knows, correct it, export it, silence it, or delete it entirely. For minors, ownership is shared with parents or guardians under age-appropriate, legally compliant arrangements.
- **P2: Consent as the only channel.** No institutional agent can read from or write to a person's PMA except through an explicit, scoped, time-limited, revocable consent grant. There is no ambient monitoring path.
- **P3: Data minimization and locality.** Personal data lives in a local-first personal vault; institutions receive derived, minimal disclosures (e.g., “homework completed: yes”) rather than raw streams (e.g., video, transcripts, biometrics).
- **P4: Human-in-the-loop escalation.** For safety-critical events (credible risk of serious harm), the system routes to designated humans: parents, counselors, clinicians, or emergency services as required by applicable law: never to automated enforcement.

- **P5: Contestability and transparency.** Every institutional query, every disclosure, and every recommendation is logged in a form the individual can review, and every automated decision affecting the person can be appealed to a human.

3. Related Work

3.1 Multi-Agent Systems and LLM Agents

Classical multi-agent systems research established foundations for autonomous, communicating agents [7], [8], including agent communication languages such as FIPA-ACL. Recent work has revisited these ideas with LLMs as the reasoning core: generative agents exhibiting believable long-horizon behavior [1], frameworks for multi-agent LLM conversation and cooperation such as AutoGen [2], and emerging interoperability standards, notably the Model Context Protocol for tool and data access. Our framework builds on this line but adds a first-class consent layer as a protocol-level primitive rather than an application-level afterthought.

3.2 Intelligent Tutoring and AI Mentorship

Intelligent tutoring systems have long demonstrated that adaptive, individualized instruction improves learning outcomes [6], [9], and LLM-based tutors such as Khan Academy's Khanmigo have brought conversational tutoring to mainstream classrooms. These systems, however, are typically bounded to a single subject, platform, or institution. The PMA generalizes the tutor into a lifelong, cross-domain mentor owned by the learner rather than the platform.

3.3 Smart Environments and Ambient Intelligence

Research on smart homes, ambient-assisted living, and smart cities has produced sensing and automation infrastructure across domains [10], [11], and commercial ecosystems (e.g., smart speakers and home platforms) demonstrate consumer appetite for ambient assistance. Critiques of these systems center on data centralization, opacity, and surveillance creep [12]. Our contribution is to accept the critique as a design input: the architecture is derived from the requirement that surveillance be structurally impossible, not merely policy-discouraged.

3.4 Privacy, Consent, and Data Governance

Privacy-by-design principles, data-minimization requirements in regulations such as the GDPR, and technical mechanisms including local-first software, differential privacy, and verifiable audit logs provide the governance toolkit for our consent layer [12], [13]. Emerging work on machine-readable consent and personal data stores (e.g., the Solid project's pod model) is closely aligned; we adapt these ideas to agent-to-agent interaction, where the consumer of data is itself an AI system acting for an institution.

3.5 Gap

To our knowledge, no prior framework combines (i) a lifelong personal LLM agent owned by the individual, (ii) institutional agents constrained to consent-scoped interaction, and (iii) human-in-the-loop

escalation as a protocol rule, all within a single, general architecture spanning education, health, home, and work. This paper specifies that architecture and a research agenda for evaluating it.

4. Two Architectures: Surveillance vs. Consent

4.1 The Surveillance Architecture and Its Failure Modes

The intuitive design for society-scale AI assistance is institutional and top-down: pervasive cameras and sensors, a central institutional AI per organization that observes everyone within it, mandatory personal agents that report upward, and automated escalation to authorities. We argue this design fails on four grounds:

- **Trust collapse.** Disclosure under compulsion is not disclosure; it is performance. Children and adults who know they are always watched learn concealment, not openness, inverting outcome O1. The therapeutic and mentoring literatures consistently identify voluntary trust as the precondition for honest disclosure and behavior change [3].
- **Wrong causal target.** Addiction and self-destructive behavior are driven primarily by pain and disconnection rather than information deficits [4], [5]. Monitoring intensity does not touch the cause; a chosen, trusted companion can.
- **Technical unreliability at the point of maximum harm.** Emotion recognition from facial expression is scientifically contested and unreliable across contexts and cultures [14], and automated dispatch of enforcement based on machine perception produces false positives whose costs fall on real people. Building enforcement on this foundation is unsafe engineering.
- **Governance risk.** An infrastructure of total observation, once built, is available to every future controller of it, benevolent or not. Systems should be designed so that abuse is structurally difficult, not merely presently unintended.

4.2 The Consent-First Architecture

The consent-first architecture inverts the direction of authority. The individual's PMA is the hub of their data and their goals; institutions are spokes that must ask. Crucially, this is not merely the ethical option: it is the effective one. Every mechanism by which the system produces its target outcomes (confiding, motivation, habit formation, mentorship) runs on trust, and trust is exactly what individual ownership and revocable consent create. Consent is therefore the load-bearing component of the design, not a regulatory tax on it. Table 1 contrasts the two architectures along the dimensions that matter for the target outcomes.

Table 1. Surveillance architecture versus consent-first architecture.

Dimension	Surveillance Architecture	Consent-First Architecture
Root of authority	Institution observes the person by default	Person owns their agent and data; institutions must ask
Participation	Mandatory; refusal penalized	Voluntary; refusal never penalized

Dimension	Surveillance Architecture	Consent-First Architecture
Data flow	Raw streams (video, audio, biometrics) to institutional cloud	Derived facts only, computed inside the personal vault
Disclosure driver	Compulsion, producing concealment and performance	Trust, producing honest confiding
Escalation	Automated dispatch to enforcement	Routed to accountable humans only
Abuse potential	Structural: total-observation infrastructure available to any future controller	Structurally limited: no ambient channel exists to abuse
Auditability	Opaque to the observed person	Every disclosure logged, human-readable, contestable

5. Proposed Architecture

5.1 Overview

The ecosystem comprises four elements: (1) Personal Mentor Agents, one per individual; (2) Institutional Coordinator Agents, one or more per organization; (3) a Consent and Disclosure Protocol governing all PMA-ICA interaction; and (4) a Safety Escalation Layer routing rare, high-severity events to humans. Computation follows a local-first, edge-cloud hybrid pattern: personal data and day-to-day reasoning stay as close to the individual's devices as feasible, with cloud resources used for heavy model inference under encryption and minimization constraints. Figure 1 shows the overall architecture and its trust boundaries.

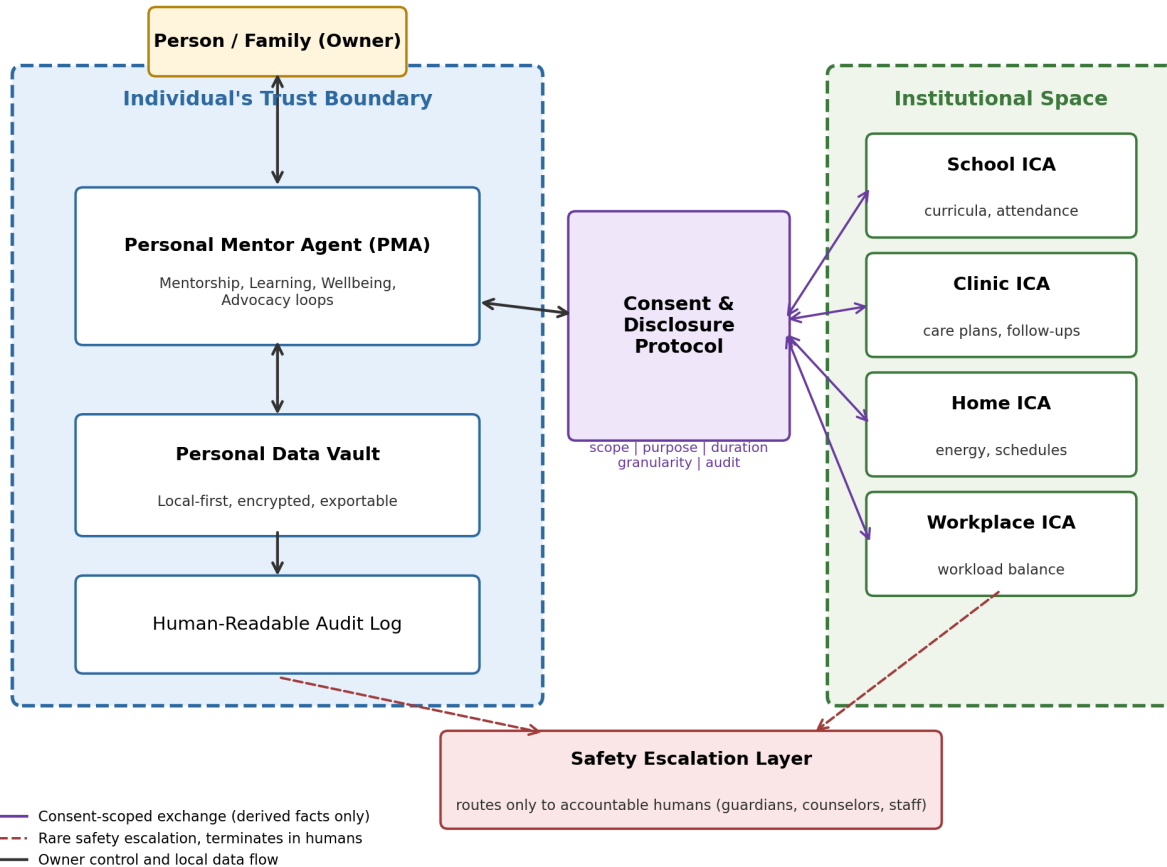


Figure 1. Reference architecture. The individual's trust boundary (left) contains the PMA, the personal data vault, and the audit log; institutions (right) interact only through the consent and disclosure protocol, and safety events terminate in accountable humans.

5.2 The Personal Mentor Agent (PMA)

The PMA is a long-lived agent instance bound to one person. Its core loops are:

- **Mentorship loop.** Daily, opt-in check-ins and conversation: goals, mood, plans, reflection. The PMA maintains a longitudinal memory (in the personal vault) of the person's interests, struggles, and progress, enabling continuity no stateless assistant can offer.
- **Learning loop.** Curriculum construction around the person's chosen interests; daily tasks with spaced repetition and difficulty adaptation; progress tracking toward mastery. The person (or, for a child, the family) chooses the domains; the PMA supplies the consistency.
- **Wellbeing loop.** Habit scaffolding for sleep, nutrition, movement, and routine, at an intensity the person selects, ranging from silent tracking to active coaching. The PMA may notice patterns (e.g., disrupted sleep, withdrawn tone) and gently ask, and the person may always decline to answer, with no record of refusal shared with anyone.

- **Advocacy loop.** The PMA represents the person to institutions: negotiating scheduling, retrieving assignments, summarizing what the person chooses to share. It is the person's agent in both the technical and the fiduciary sense; its duty of loyalty runs to the individual alone.

5.3 The Institutional Coordinator Agent (ICA)

An ICA manages an institution's legitimate coordination functions: a school ICA tracks curricula, assignments, attendance, and facility safety; a clinic ICA manages appointments, care plans, and follow-ups; a workplace ICA handles scheduling, workload balance, and training. ICAs observe *institutional* processes and spaces consistent with law and posted policy (e.g., a camera at a school entrance used for automated attendance, with notice and lawful basis), but they hold no channel into any individual's personal life except through the consent protocol. An ICA may *ask* a student's PMA whether assigned work was completed; it cannot read the student's conversations, mood history, or vault.

5.4 The Consent and Disclosure Protocol

All PMA-ICA interaction is mediated by consent grants with five mandatory fields: **scope** (exactly which derived facts may be disclosed), **purpose** (the institutional use), **duration** (expiry, renewable only by the person), **granularity** (derived summaries by default; raw data only by separate, elevated grant), and **audit** (every exercised grant is logged to the person's vault in human-readable form). Grants are revocable at any time with immediate effect. For minors, grants require guardian co-signature, with disclosure scopes constrained by child-protection law and by age-appropriate design codes. The protocol composes naturally with emerging agent-interoperability standards (e.g., tool-use protocols), adding consent as a first-class message type rather than an out-of-band legal document. Figure 2 illustrates the full lifecycle of a grant, from request through runtime exercise and audit.

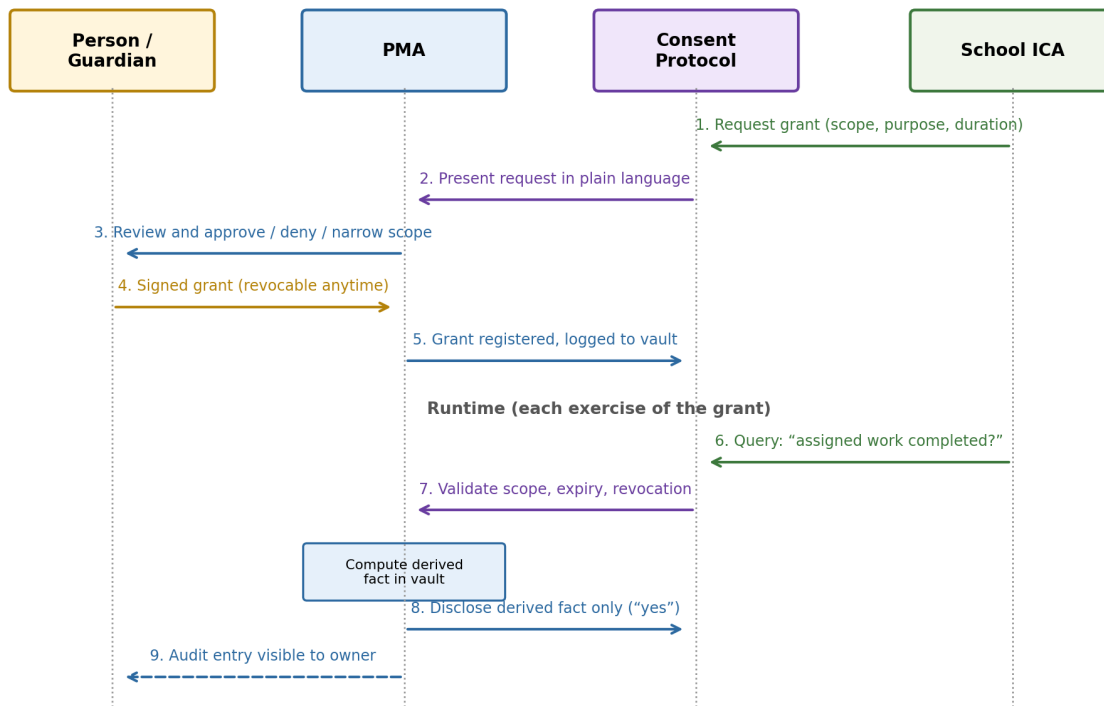


Figure 2. Consent workflow. Steps 1 through 5 establish a scoped, revocable grant with the owner's explicit approval; steps 6 through 9 show a runtime query answered with a derived fact and logged to the owner's audit trail.

5.5 Safety Escalation Layer

Two narrow classes of events bypass ordinary consent scoping, both terminating in humans rather than automation: (1) **imminent-harm events**, where the PMA has strong evidence of immediate danger to the person or others, triggering notification to pre-designated humans (guardians, counselors, emergency contacts) under rules the person or family configured in advance and consistent with mandatory-reporting law; and (2) **physical-safety events in institutional space**, e.g., a school ICA's camera detecting violence in a hallway, which alerts on-site staff (trained humans who can assess context) rather than dispatching enforcement automatically. In both classes the system's role ends at informing an accountable human; it never adjudicates or enforces.

5.6 Data Architecture

Each person's vault is local-first: an encrypted store on personal devices with optional end-to-end-encrypted backup. Institutional queries are answered by computing the minimal derived fact inside the person's trust boundary and disclosing only that fact. Aggregate institutional analytics (e.g., "how is the class progressing?") are served through privacy-preserving aggregation so that no individual's data is exposed. This inverts the conventional smart-environment pattern in which raw sensor streams flow to institutional clouds by default.

6. Illustrative Scenarios

6.1 Education

A twelve-year-old's family activates a learning plan around her interest in electronics. Her PMA builds a curriculum, assigns a small daily task, and adapts difficulty to her progress. The school's ICA queries her PMA nightly under a standing grant (“assigned work completed?”), receiving yes/no, nothing more. One week her PMA notices her engagement drop and her tone flatten; it gently asks whether something is wrong. She confides a bullying incident. With her agreement, the PMA helps her tell her parents and requests a counselor appointment through the school ICA, disclosing only “student requests counselor time,” not the content. The adults take it from there. Every element of outcome O1 is served; no surveillance occurred.

6.2 Healthcare

An adult managing prediabetes grants his clinic's ICA a six-month scope: weekly averages of steps, sleep hours, and self-reported meals: derived numbers, not raw logs. His PMA handles the daily coaching he asked for. When his numbers drift, the clinic ICA flags his care team, and a human clinician calls him. He revokes the grant after the program ends; disclosure stops that instant.

6.3 Home

A household's home ICA optimizes energy and schedules around preferences each member's PMA shares under family-configured scopes. An elderly parent opts into fall-detection sensing in her own home, with alerts routed to her daughter. It is her choice of watcher, her scope, her revocation right. Assistance reaches the person who wants it, configured by the person who receives it.

6.4 Workplace

A company's ICA balances workload using capacity signals employees' PMAs share under a collectively negotiated scope: available/at-capacity, never keystrokes, never sentiment. An employee's PMA, loyal to him and not to the employer, notices a pattern of late-night work and rising stress and suggests, privately, that he use the counseling benefit. The employer never learns of the suggestion.

6.5 Cross-Domain Synergy Without Cross-Domain Surveillance

Note what makes these scenarios compose: the *person* is the integration point. Health context can inform learning pace, and home context can inform work capacity, because all of it lives in one vault under one owner, the individual, who decides which derived slivers each institution sees. Cross-domain benefit, the central promise of unified smart environments, is achieved precisely without a central observer.

7. Proposed Evaluation Plan

This is a framework paper; we report no empirical results. Instead we specify how the framework's claims should be tested, and we invite replication and critique of the plan itself.

7.1 Hypotheses

- **H1 (Disclosure):** Users of an owned, consent-scoped PMA will disclose sensitive difficulties (stress, conflict, struggling in a subject) at higher rates and earlier than users of an institutionally provided monitoring assistant.
- **H2 (Learning consistency):** Learners with a PMA running spaced daily tasks will show higher 8-week retention and lower topic-abandonment than self-directed controls, comparable to bounded ITS gains reported in prior literature.
- **H3 (Minimal disclosure sufficiency):** Institutional coordination tasks (attendance, assignment tracking, scheduling, workload balancing) can be completed with derived-fact disclosure alone, with no statistically significant loss of coordination quality versus raw-data baselines.
- **H4 (Trust and adoption):** Perceived control (measured by standardized privacy-control scales) will predict sustained voluntary engagement with the PMA at 3 and 6 months.

7.2 Study Designs and Metrics

H1 and H4 call for randomized comparisons between ownership framings with validated trust and disclosure instruments; H2 calls for a controlled learning study with pre-registered outcomes (retention tests, streak persistence, abandonment rates); H3 can be evaluated in simulation first, by implementing PMA/ICA message exchange over standard agent frameworks and measuring task-completion quality, message overhead, and latency under derived-fact versus raw-data policies, followed by small field pilots. All human-subject studies require ethics-board approval, informed consent, and, for minors, guardian consent and age-appropriate protocols. We commit to reporting negative results. Table 2 summarizes the plan.

Table 2. Evaluation plan: hypotheses, study designs, and primary metrics.

Hyp.	Claim	Study Design	Primary Metrics
H1	Owned, consent-scoped PMAs increase early disclosure of difficulties versus institutional monitoring assistants	Randomized comparison of ownership framings; ethics-approved; validated instruments	Disclosure rate and latency; trust scale scores
H2	PMA-driven spaced daily tasks improve retention and reduce topic abandonment	Controlled 8-week learning study; pre-registered outcomes	Retention test scores; streak persistence; abandonment rate
H3	Derived-fact disclosure suffices for institutional coordination	Agent simulation (PMA/ICA message exchange), then small field pilots	Task-completion quality; message overhead; latency
H4	Perceived control predicts sustained voluntary engagement	Longitudinal cohort at 3 and 6 months	Privacy-control scale; retention and engagement rates

7.3 Prototype Path

A minimal viable prototype requires only: an LLM-backed PMA with a local encrypted store; one ICA (e.g., a classroom coordinator); the consent-grant message schema; and an audit log viewer. The author's prior work building production LLM-agent systems and MCP-based integrations suggests this is achievable with current open tooling; a reference implementation is planned as the immediate next step and will be reported separately with real measurements.

8. Ethical Considerations and Limitations

8.1 Ethical Boundaries

The framework's legitimacy depends on boundaries it must never cross, which we state as commitments: participation is voluntary and non-participation must never be penalized by institutions; the PMA's loyalty runs to the individual and may not be silently dual-mastered; emotion inference, where used at all, is treated as a low-confidence conversational cue, never as evidence for institutional decisions; children's arrangements must follow child-protection law and give the child growing autonomy with age; and no component may automate enforcement, punishment, or denial of services. A consent architecture wrapped around coercive deployment (e.g., “consent” required to attend school) would reproduce the surveillance architecture under another name; deployment policy is therefore part of the safety case, not external to it.

8.2 Risks

We identify four risks requiring ongoing research: **dependency**: a mentor always available may crowd out human relationships; the PMA must be designed to route people toward humans, not substitute for them; **manipulation**: a trusted lifelong agent is a high-value target for commercial or political influence, arguing for strict fiduciary rules and open auditing of PMA objectives; **inequality**: if PMAs are premium goods, they amplify rather than reduce gaps, arguing for public-infrastructure or open-source provision; and **model failure**: LLM errors in sensitive conversations (health, distress) can cause harm, requiring conservative escalation design and human referral defaults.

8.3 Limitations

This paper is conceptual: its claims about effectiveness are grounded in prior literature and design analysis, not in new experiments. The consent protocol is specified at the message-schema level, not yet as a formal standard with security proofs. Legal analysis across jurisdictions (data protection, education, health, and child-safety law) is out of scope and required before deployment. These are the subjects of the evaluation plan in Section 7.

9. Conclusion

The wish behind pervasive smart environments is humane: that no child's pain go unnoticed, that no one drift into self-destruction for lack of connection, that everyone have a patient mentor and a path to mastery and dignity. This paper argues that the architecture usually imagined for that wish, universal observation under institutional authority, would betray it, and that the same outcomes are better served by inverting the design: give every person an agent of their own, make consent the only channel between

personal and institutional AI, and keep humans in the loop wherever stakes are high. The result preserves the full ambition (cross-domain, lifelong, proactive support) while making abuse structurally difficult rather than merely discouraged. The next step is empirical: building the reference prototype and running the studies specified here. We offer this framework as a starting point and invite the community to test, break, and improve it.

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